



BOILEAU GUILLAUME

Space Systems Engineer | RAMS, Safety, Reliability, Risk Analysis & Validation

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EXPERIENCE

Software Engineer – System Integration, Operational Reliability & Hardware Interfaces

VEV Platform Services France

📅 2025 – 2026

📍 Valbonne, France

- Developed and maintained web and backend services for an EV charging CPMS platform using **TypeScript, Angular and Node.js**.
- Contributed to the integration of software services connected to operational charging infrastructure and hardware systems.
- Supported the analysis of hardware/software interactions, operational data exchanges and service behaviours in production.
- Investigated incidents, abnormal behaviours, communication issues and data inconsistencies affecting service continuity.
- Contributed to monitoring, verification and correction of operational data flows, including FTP-based exchanges.
- Supported diagnosis, prioritization and issue follow-up with technical stakeholders in an operational industrial context.
- Strengthened experience with reliability constraints, maintainability and service robustness in software systems connected to field equipment.

Senior R&D Engineer – Space Systems, Validation, Risk Analysis & Technical Coordination

Laboratoire Lagrange, Observatoire de la Côte d'Azur, CNES

📅 2023 – 2025

📍 Nice, France

- Contributed to engineering and analysis activities for the **LISA ESA/NASA space mission**, in an international environment involving complex mission-level requirements, scientific performance constraints and high technical rigor.
- Developed **CATpyLISA**, a Python platform for large-scale model integration, comparison, analysis and validation workflows.
- Structured validation workflows to compare outputs, verify expected behaviours, identify deviations and consolidate confidence in technical results.
- Worked at the interface between scientific requirements, software development, data processing, modelling assumptions and mission-level performance objectives.
- Identified inconsistencies, limitations, uncertainty sources and critical behaviours in complex software and analysis chains.
- Produced risk-oriented technical analyses by evaluating the impact of modelling choices, numerical behaviours and data-processing assumptions on final results.
- Supported traceability of configurations, parameters, assumptions, validation results and technical conclusions.
- Contributed to technical review preparation through documentation of validation procedures, limitations, assumptions and interpretation of results.
- Coordinated technical activities involving contributors, research engineers, technicians and interns.
- Supervised Master's thesis interns on software, data analysis, validation and modelling tasks.
- Operated in a collaborative international framework requiring technical English, structured reporting, versioned workflows and rigorous documentation.

Data Scientist – Instrument Performance, Noise Modelling & Failure Diagnostics

Universiteit Antwerpen, Belgium

📅 2021 – 2023

📍 Antwerp, Belgium

- Analysed instrumental and environmental effects impacting the sensitivity, stability and performance of precision measurement systems.
- Developed Python-based analysis pipelines for noise source identification, uncertainty estimation, Bayesian inference and statistical modelling.
- Investigated unexpected behaviours, degradation patterns and correlations between environmental conditions and measurement performance.
- Supported technical interpretation of system limitations by identifying dominant contributors, weak points and possible mitigation directions.
- Produced reproducible and traceable technical analyses in a collaborative international environment.
- Contributed to studies connected to gravitational-wave detector systems, where performance, noise control, robustness and system-level sensitivity are critical.
- Communicated complex technical results to scientific and engineering stakeholders through structured reporting and collaborative review.

Junior R&D Engineer – Signal Analysis, Simulation & System Performance Assessment

ARTEMIS, Observatoire de la Côte d'Azur

📅 2018 – 2021

📍 Nice, France

SUMMARY AND STRENGTHS

Space systems and software engineer with strong experience in complex system analysis, reliability-oriented validation, risk identification, performance assessment and technical diagnostics. Background developed through major international collaborations including **LISA**, **LIGO-Virgo-KAGRA** and **Einstein Telescope**, complemented by recent industrial software experience involving hardware/software interfaces and operational reliability.

Experienced in **requirements-oriented validation**, **risk analysis**, **failure diagnostics**, **anomaly investigation**, **model comparison**, **uncertainty analysis**, **software integration**, **traceability**, **technical documentation**, **Python automation** and **international technical coordination**.

Able to analyse complex technologies and systems, identify critical behaviours, consolidate technical evidence, document assumptions and support design or mitigation decisions in high-constraint environments.

Relevant for **RAMS**, **Safety**, **Reliability**, **Availability**, **Maintainability**, **space systems engineering**, **system verification**, **risk mitigation**, **technical reviews** and **complex system validation**.

FIT FOR THALES RAMS / SAFETY

- Experience with complex space and precision-instrument systems requiring robust modelling, validation and performance assessment.
- Ability to identify technical limitations, uncertainty sources, critical behaviours and residual risks.
- Strong background in verification, validation, traceability, result consolidation and technical documentation.
- Experience in anomaly investigation, root-cause-oriented analysis and diagnostic reporting.
- Familiar with system-level constraints, mission-level performance, reliability of analysis chains and robustness of technical workflows.
- Comfortable working at the interface between requirements, modelling, software, data, system constraints and operational needs.
- Relevant background for RAMS activities involving reliability, availability, maintainability, safety, risk analysis and conformity evidence.
- Used to collaborative and transnational environments requiring technical English, structured communication and high execution rigor.

TEAM MANAGEMENT

- Supervision of **3 Master's thesis interns (M2)** during engineering, modelling and software-development phases.
- Coordination of technical activities involving **research engineers and technicians** on a **data processing pipeline** for the **LISA space mission**.
- Experience in organizing tasks, supporting contributors, reviewing outputs and maintaining alignment across collaborative technical developments.

LANGUAGES

English



Spanish



EDUCATION

PhD in Physics

- Conducted research and engineering-oriented analysis for the preparation of gravitational-wave space missions, with a strong focus on **LISA**.
- Developed methods for the separation of weak overlapping signals in complex low-SNR datasets.
- Built simulation and analysis workflows to evaluate system performance under realistic noise, foreground and uncertainty conditions.
- Compared simulated outputs, model behaviours and analysis results to assess consistency, robustness and performance limitations.
- Investigated noise sources and correlations through diagnostic analyses relevant to precision instruments.
- Produced peer-reviewed results contributing to the understanding of performance limits for future gravitational-wave space missions.
- Developed strong expertise in modelling, uncertainty analysis, signal processing, technical validation and scientific software workflows.

Université Côte d'Azur

📅 2018 – 2021

📍 Nice, France

Selective Graduate Program in Fundamental Physics

Université Paris-Saclay

📅 2016 – 2018

📍 Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

📅 2017 – 2018

📍 Paris, France

BSc in Fundamental Physics

Université Paris-Saclay

📅 2015 – 2016

📍 Orsay, France

PROFESSIONAL TRAINING

Supervised Machine Learning

DeepLearning.AI – Andrew Ng

📅 2020

📍 Coursera

PROJECTS

Co-author and full member of the LISA Consortium

ESA/NASA Space Mission – Large-scale International Collaboration

📅 2018 – now

Contribution to software, analysis and validation workflows for the LISA space mission within an international engineering and scientific collaboration involving ESA, NASA and multiple research institutes.

Role: Contribution to software workflows, technical coordination, complex signal-processing activities, model integration, validation campaigns, performance assessment and collaborative engineering tasks.

RAMS relevance: Work involving traceability, consistency checks, performance limitations, uncertainty analysis, model comparison, validation evidence and identification of critical behaviours in complex mission-related workflows.

Project relevance: Work at the interface between software development, scientific requirements, data processing, validation workflows, technical contributors and international coordination.

Program scale: Major international space mission with an estimated budget of approximately 2 billion EUR over 10 years.

Co-author and full member of Einstein Telescope and LIGO–Virgo–KAGRA Collaborations

International Collaborations

📅 2021 – now

Contribution to collaborative developments in data analysis, signal characterization, software methods and technical investigations within the LIGO–Virgo–KAGRA and Einstein Telescope collaborations.

Role: Development of analysis tools, contribution to technical activities and support to complex software workflows in high-standard collaborative environments.

System impact: Contributed to studies on environmental and instrumental noise sources, helping identify fundamental design constraints and performance limitations for next-generation gravitational-wave observatories.

Environment: Large-scale international collaborations involving strong technical requirements, structured methods, technical review, versioned software workflows and coordinated contributions.

STRENGTHS

RAMS, Safety & risk analysis

RAMS Safety Reliability Availability Maintainability Risk Analysis Failure Diagnostics Failure Mode Awareness Criticality Analysis Residual Risk Analysis Risk Mitigation Support System Robustness Performance Consolidation Uncertainty Analysis Compliance Mindset Safety-Critical Awareness

Systems & validation engineering

Systems Engineering Space Systems Satellite Mission Context Mission-Level Performance Requirements-Oriented Validation Verification & Validation System Validation Validation Campaigns Test Strategy Support Software IVVQ Software Integration Integration Testing Requirements Traceability Anomaly Analysis Root Cause Analysis Technical Diagnostics Issue Resolution Hardware / Software Interfaces EGSE-Relevant Interfaces

Technical coordination

Technical Project Coordination Work Package Structuring Task Follow-Up Contributor Alignment Technical Roadmap Support Delivery Support Technical Reporting Stakeholder Communication Cross-Functional Coordination Team Coordination Issue Follow-Up Decision Support Documentation Quality Structured Execution

Technical skills

Python Shell Scripting Bash Linux Git GitLab Docker CI/CD Test Automation Scripting SQL TypeScript Angular Node.js C Matlab
Pandas REST API FTP Issue Tracking Technical Reporting Condor Slurm PyTorch MongoDB Mathematica

Domain knowledge

Space Systems Satellite Systems ESA/NASA Collaboration Transnational Environment Scientific Payload Analysis Precision Instrumentation Space Mission Data Proc
Large-Scale International Projects Critical Systems Awareness Industrial Software Technical English

Soft skills

Autonomy Analytical Thinking Execution Rigor Problem Solving Adaptability Responsibility Attention to Detail Collaboration
Structured Communication Customer Awareness Technical Interface Leadership Potential
Applications: RAMS, Safety, reliability engineering, space systems, system validation, software IVVQ, risk analysis and complex system performance assessment